GPT-OSS 20B MoE Benchmark Report

Primus + Megatron-LM on 8× NVIDIA H200

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Objective

20B MoE pretraining benchmark on 8× H200 GPUs with MLPerf target loss (3.34). Focus on stable distributed training and time-to-convergence.

System Configuration

Hardware: 8× H200 (141GB)
Framework: Primus + Megatron-LM
Config: Micro=1, Global=16, Seq=8192
Note: Micro-batch reduced (2→1) due to memory limits.

Challenges

Patch mismatch fixed via commit pinning.
OOM (134GB) resolved using micro-batch reduction and CUDA memory tuning.

Time-to-Convergence

Target Loss: 3.34
Achieved: 3.325
Time: 6.09 hours

Result: Faster than reference (6.5h) despite reduced memory.

Training Dynamics

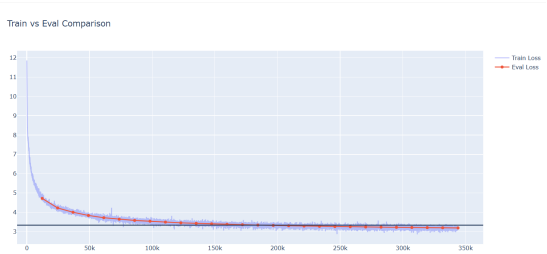

Figure 1: Train vs Eval Loss

Eval loss provides stable convergence signal; training loss remains noisy.

Results

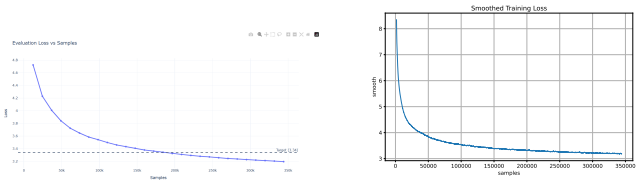

Figure 2: Eval vs Training Loss

Throughput

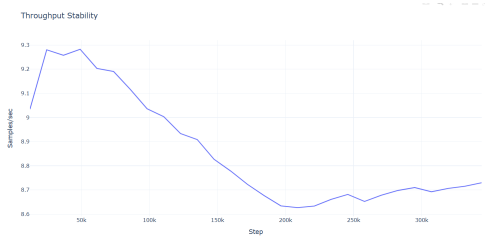

Figure 3: Throughput Stability

Stable throughput: ∼8.7 samples/sec.

GPU Utilization

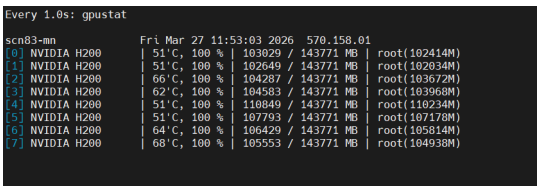

Figure 4: GPU Utilization

All GPUs operate near 100% utilization with balanced memory usage.

Insights

- Time-to-target is the key MLPerf metric
- Eval loss governs convergence
- Micro-batch impacts memory efficiency
- MoE increases memory and communication overhead

Conclusion

Achieved MLPerf target in **6.09 hours** with stable distributed training, high GPU utilization, and efficient memory usage.